Lab 4

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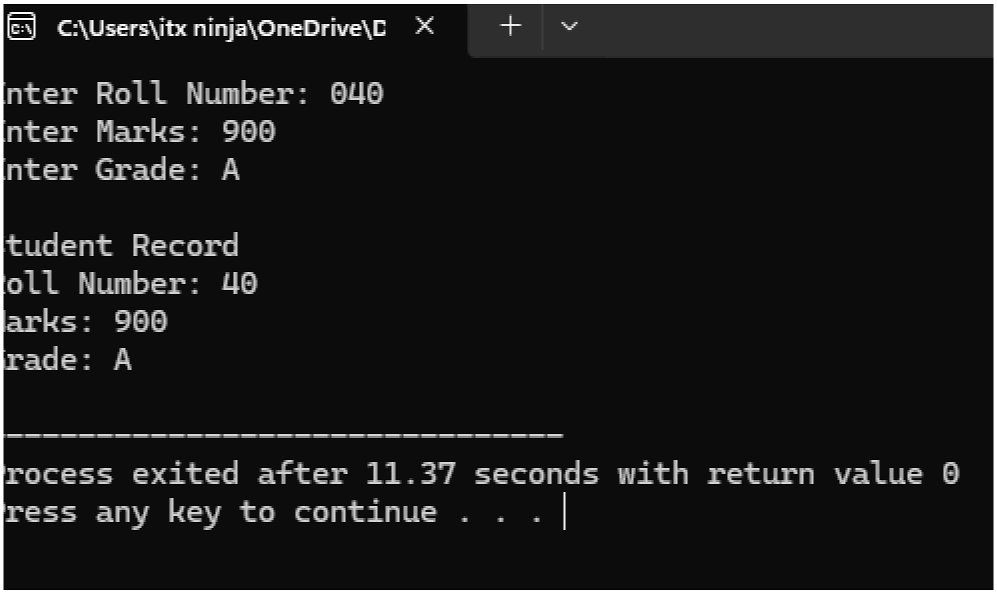
Task1:

.cpp

```cpp
#include <iostream>
using namespace std;
struct Result
{
    int marks;
    char grade;
};
struct Record
{
    int rollNo;
    Result r;
};
int main()
{
    Record student;
    cout << "Enter Roll Number: ";
    cin >> student.rollNo;
    cout << "Enter Marks: ";
    cin >> student.r.marks;
    cout << "Enter Grade: ";
    cin >> student.r.grade;
    cout << "\nStudent Record" << endl;
    cout << "Roll Number: " << student.rollNo << endl;
    cout << "Marks: " << student.r.marks << endl;
    cout << "Grade: " << student.r.grade << endl;
    return 0;
}
```


C:\Users\itx ninja\OneDrive\D
inter Roll Number: 040
inter Marks: 900
inter Grade: A
tudent Record
oll Number: 40
larks: 900
rade: A
----
rocess exited after 11.37 seconds with return value 0
ress any key to continue . . .

Task2:

```
/*task2
#include <iostream>
using namespace std;
struct Data
{
    string name;
    string color;
};
struct Car
{
    int model;
    int topSpeed;
    int gears;
    Data d;
};
int main()
{
    Car c;
    cout << "Enter Car Name: ";
    cin >> c.d.name;
    cout << "Enter Car Color: ";
    cin >> c.d.color;
    cout << "Enter Model: ";
    cin >> c.model;
    cout << "Enter Top Speed: ";
    cin >> c.topSpeed;
    cout << "Enter Number of Gears: ";
    cin >> c.gears;

    if (c.topSpeed > 200 && c.gears > 4)
    {
        cout << "\nCar Details" << endl;
        cout << "Name: " << c.d.name << endl;
        cout << "Color: " << c.d.color << endl;
        cout << "Model: " << c.model << endl;
```

```
C:\Users\itx ninja\OneDrive\D

Enter Car Name: BMWi7
Enter Car Color: black
Enter Model: 2026
Enter Top Speed: 500
Enter Number of Gears: 4

Your car is not suitable for the race.

--------------------------------
Process exited after 30.11 seconds with return value 0
Press any key to continue . . .
```

Task3:

```
*task3
include <iostream>
sing namespace std;
truct Complex

    float real;
    float imaginary;
;
oid addition(Complex n1, Complex n2)

    cout << "\nAddition = "
         << n1.real + n2.real << " + "
         << n1.imaginary + n2.imaginary << "i" << enc

oid subtract(Complex n1, Complex n2)

    cout << "Subtraction = "
         << n1.real - n2.real << " + "
         << n1.imaginary - n2.imaginary << "i" << enc

oid multiply(Complex n1, Complex n2)

    float realPart, imagPart;
    realPart = (n1.real * n2.real) - (n1.imaginary *
    imagPart = (n1.real * n2.imaginary) + (n1.imaginc
    cout << "Multiplication = "
         << realPart << " + "
         << imagPart << "i" << endl;

nt main()

    Complex num1, num2;
    cout << "Enter real part of first number: ";
```

```
C:\Users\itx ninja\OneDrive\D
Enter real part of first number: 2
Enter imaginary part of first number: 0
Enter real part of second number: 4
Enter imaginary part of second number: 1

Addition = 6 + 1i
Subtraction = -2 + -1i
Multiplication = 8 + 2i

--------------------------------
Process exited after 16.5 seconds with return value 0
Press any key to continue . . .
```

Task 4:

```
C:\Users\itx ninja\OneDrive\D

Enter details of Book 1
Enter Title: the art of letting go
Enter ISBN: Enter Price: Enter Publication Year: Enter Author Name: Enter Author Nationality:
Enter details of Book 2
Enter Title: Enter ISBN: Enter Price: Enter Publication Year: Enter Author Name: Enter Author Nationality:
Enter details of Book 3
Enter Title: Enter ISBN: Enter Price: Enter Publication Year: Enter Author Name: Enter Author Nationality:
Books Published After 2015

--------------------------------
Process exited after 11.67 seconds with return value 0
Press any key to continue . . .
```

```
*task4
include <iostream>
include <string>
sing namespace std;
truct Author

    string name;
    string nationality;
;
truct Book

    string title;
    string ISBN;
    double price;
    int publicationYear;
    Author a;
;
nt main()

    Book b[3];
    for (int i = 0; i < 3; i++)
    {
        cout << "\nEnter details of Book " << i + 1 << endl;
        cout << "Enter Title: ";
        cin >> b[i].title;
        cout << "Enter ISBN: ";
        cin >> b[i].ISBN;
        cout << "Enter Price: ";
        cin >> b[i].price;
        cout << "Enter Publication Year: ";
        cin >> b[i].publicationYear;
        cout << "Enter Author Name: ";
        cin >> b[i].a.name;
        cout << "Enter Author Nationality: ";
```

Task 5:

```
/*task5
#include <iostream>
#include <string>
using namespace std;
struct Student
{
    string name;
    int rollNo;
    float marks[3];
    float gpa;
};
void calculateGPA(Student &s)
{
    float average;
    average = (s.marks[0] + s.marks[1] + s.marks[2]) / 3;
    s.gpa = average / 10;

    if (s.gpa > 4.0)
    {
        s.gpa = 4.0;
    }
}
void displayStudent(Student s)
{
    cout << "\nStudent Details" << endl;
    cout << "Name: " << s.name << endl;
    cout << "Roll Number: " << s.rollNo << endl;
    cout << "Marks Subject 1: " << s.marks[0] << endl;
    cout << "Marks Subject 2: " << s.marks[1] << endl;
    cout << "Marks Subject 3: " << s.marks[2] << endl;
    cout << "GPA: " << s.gpa << endl;
}
int main()
{
    Student s[2];
```

```
C:\Users\itx ninja\OneDrive\D
Enter Marks of Subject 1: 95
Enter Marks of Subject 2: 98
Enter Marks of Subject 3: 99

Enter details of Student 2
Enter Name: shahzaib
Enter Roll Number: 018
Enter Marks of Subject 1: 98
Enter Marks of Subject 2: 34
Enter Marks of Subject 3: 99

Student Details
Name: Qadeer
Roll Number: 40
Marks Subject 1: 95
Marks Subject 2: 98
Marks Subject 3: 99
GPA: 4

Student Details
Name: shahzaib
Roll Number: 18
Marks Subject 1: 98
Marks Subject 2: 34
Marks Subject 3: 99
GPA: 4

--------------------------------
Process exited after 40.02 seconds with return value 0
Press any key to continue . . .
```

Task 6:

```
C:\Users\itx ninja\OneDrive\D

Enter Account Number: 03264543916
Enter Holder Name: qadeer durrani
Enter Initial Balance:
Enter amount to deposit:
Deposit Receipt
Deposited Amount: 2.32211e-322
Current Balance: 2.32211e-322

Enter amount to withdraw:
Withdrawal Successful
Remaining Balance: 0

--------------------------------
Process exited after 20.36 seconds with return value 0
Press any key to continue . . .
```

```
/*task6
#include <iostream>
#include <string>
using namespace std;
struct Account
{
    string accountNumber;
    string holderName;
    double balance;
};
Account createAccount()
{
    Account a;
    cout << "Enter Account Number: ";
    cin >> a.accountNumber;
    cout << "Enter Holder Name: ";
    cin >> a.holderName;
    cout << "Enter Initial Balance: ";
    cin >> a.balance;
    return a;
}

void deposit(Account &a, double amount)
{
    a.balance = a.balance + amount;
    cout << "\nDeposit Receipt" << endl;
    cout << "Deposited Amount: " << amount << endl;
    cout << "Current Balance: " << a.balance << endl;
}
bool withdraw(Account &a, double amount)
{
    if (amount <= a.balance)
    {
        a.balance = a.balance - amount;
        return true;
```

Home task

Task1 :

```cpp
#include <string>
using namespace std;
struct Date
{
    int day;
    int month;
    int year;
};
struct Phonebook
{
    string name;
    string city;
    string phoneNumber;
    Date d;
};
int main()
{
    Phonebook p;
    cout << "Enter Name: ";
    cin >> p.name;
    cout << "Enter City: ";
    cin >> p.city;
    cout << "Enter Phone Number: ";
    cin >> p.phoneNumber;
    cout << "Enter Day: ";
    cin >> p.d.day;
    cout << "Enter Month: ";
    cin >> p.d.month;
    cout << "Enter Year: ";
    cin >> p.d.year;
    cout << "\nPhonebook Details" << endl;
    cout << "Name: " << p.name << endl;
    cout << "City: " << p.city << endl;
    cout << "Phone Number: " << p.phoneNumber << endl;
    cout << "Date: " << p.d.day << "/" << p.d.month << "/" << p.d.year << endl;
```

C:\Users\itx ninja\OneDrive\D

```
Enter Name: qadeer
Enter City: lahore
Enter Phone Number: 03264543916
Enter Day: sunday
Enter Month: Enter Year:
Phonebook Details
Name: qadeer
City: lahore
Phone Number: 03264543916
Date: 0/0/1


--------------------------------
Process exited after 21.77 seconds with return value 0
Press any key to continue . . .
```

Task2 :

```
/*task2
#include <iostream>
using namespace std;
struct Parameters
{
    float length;
    float width;
};
struct Result
{
    float area;
    float perimeter;
};

struct Rectangle
{
    Parameters p;
    Result r;
};
int main()
{
    Rectangle rect;
    cout << "Enter Length: ";
    cin >> rect.p.length;
    cout << "Enter Width: ";
```

```
C:\Users\itx ninja\OneDrive\D

Enter Length: 5
Enter Width: 15

Rectangle Details
Length: 5
Width: 15
Area: 75
Perimeter: 40

--------------------------------
Process exited after 7.778 seconds with return value 0
Press any key to continue . . .
```

Task 3:

```
'task3
:nclude <iostream>
:nclude <string>
;ing namespace std;
:ruct Instructor

   string name;
   string department;
'
:ruct Course

   string courseCode;
   string courseName;
   int creditHours;
   int maxSeats;
   int enrolledStudents;
   Instructor ins;
'
ool enrollStudent(Course &c)

   if (c.enrolledStudents < c.maxSeats)
   {
       c.enrolledStudents++;
       return true;
   }
   else
   {
       return false;
   }
```

```
C:\Users\itx ninja\OneDrive\D
Enter Enrolled Students: 29
Enter Instructor Name: miss fatime
Enter Department:
Student enrolled in 029a

Student enrolled in 0310b

Course Details
Course Code: 029a
Course Name: Oops
Credit Hours: 3
Maximum Seats: 45
Enrolled Students: 41
Seats Remaining: 4
Instructor Name: miss
Department: eisha

Course Details
Course Code: 0310b
Course Name: Python
Credit Hours: 3
Maximum Seats: 39
Enrolled Students: 30
Seats Remaining: 9
Instructor Name: miss
Department: fatime
```

Task 4:

```
*/
/*task4
#include <iostream>
#include <string>
using namespace std;
struct Date
{
    int day;
    int month;
    int year;
};
struct Doctor
{
    string name;
    string specialization;
};
struct Patient
{
    string patientID;
    string name;
    int age;
    Date admissionDate;
    Doctor assignedDoctor;
    double dailyCharge;
};
double calculateBill(Patient p, int days)
{
    return p.dailyCharge * days;
}
void displayPatientReport(Patient p, int days)
{
    double totalBill = calculateBill(p, days);
    cout << "\n==============================" << endl;
    cout << "      Hospital Report" << endl;
```

C:\Users\itx ninja\OneDrive\D

```
Enter details of Patient 1
Enter Patient ID: 1
Enter Name: abdullah
Enter Age: 15
Enter Admission Day: 23
Enter Admission Month: April
Enter Admission Year: Enter Doctor Name: Enter Specialization: Enter Daily Charge: Enter Number of Days Stayed:
Enter details of Patient 2
Enter Patient ID: Enter Name: Enter Age: Enter Admission Day: Enter Admission Month: Enter Admission Year: Enter Docto
Name: Enter Specialization: Enter Daily Charge: Enter Number of Days Stayed:
==============================
      Hospital Report
==============================
Patient ID: 1
Name: abdullah
Age: 15
Admission Date: 23/0/0
Doctor Name:
Specialization:
Daily Charge: 5.73359e-317
Days Stayed: 8
Total Bill: 4.58687e-316
==============================

==============================
      Hospital Report
==============================
Patient ID:
```